

```
GENERICUS
// class
class Box<T> { }
// variable
Box<String> b = new Box<String>();
// method
public <T> void doStuff(T value) { }
```

```
JAVA BEING STUPID
T obj = new T(); // error
obj instanceof T; // error
T heh = (T) "asdf"; // no typechecking
var s = Stack<int>(); // no primitive types
private static T x; // static no workkey
Node<Number> s = new Node<Integer>; // error
class IThrowAnErrorBecauseWhyNot {
    void m(List<String> list) { }
    void m(List<Integer> list) { }
} // error because of type erasure & identifiability
```

```
ITERATOR
for (String s : stringList) { } // ==
for (Iterator<String> i = stringList.iterator();
    i.hasNext();) {
    String s = i.next();
}
```

```
ITERABLE
interface Iterable<T> {
    Iterator<T> iterator();
}
interface Iterator<E> {
    boolean hasNext();
    E next();
}
```

```
COMPARABLE
TODO:
Type signature?
```

```
static <T extends Comparable> T doStuff(T a, T b) {
    // compareTo is possible with all types
    return a.compareTo("b") > 0 ? a : b;
}
```

```
static <T extends Comparable<T>> T doStuff(T a, T b)
{
    // compareTo is possible only with T
    return a.compareTo(b) > 0 ? a : b;
}
```

```
EXAMPLE
class Graphic {
    public void draw() { }
}
class Rectangle extends Graphic { }
class Circle extends Graphic { }
class GraphicStack<T extends Graphic>
    extends Stack<T> {
    public void drawAll() {
        for (T item : this) item.draw();
    }
}
```

```
BYTECODE
TODO:
```

```
TYPE ERASURE
- Introduced because of "bAcKwArDs CoMpAtAbIlItY"
- No type information at runtime (Non-Reifiable Type)
```

```
class MyStack<T> {
    Entry<T> top;
    int push(T value) { }
}
// becomes
class MyStack {
    Entry top;
    int push(Object value) { }
}
/* with bounds */
class MyStack<T extends Comparable<T>> {
    Entry<T> top;
    int push(T value) { }
}
// becomes
class MyStack {
```

```
Entry top;
int push(Comparable value) { }
}
/* what happens at/before runtime */
MyStack<String> s = new MyStack<String>();
s.push("Eh");
String x = s.pop();
// becomes
MyStack s = new MyStack();
s.push("Eh");
String x = (String) s.pop();
```

```
WILDCARDS
void printGs(List<Graphic> gs) { }
List<Graphic> gs1 = new ArrayList<>();
printGs(gs1); // ok
List<Circle> gs2 = new ArrayList<>();
printGs(gs2); // error
/* solution */
void printGs(List<? extends Graphic> gs) { }
```

	Type	Compatible Type-Args	R	W
Invariance	C<T>	T	✓	✓
Covariance	C<? extends T>	T and sub-types	✓	✗
Contravariance	C<? super T>	T and basis-types	✗	✓
Bivariance	C<?>	All	✗	✗

```
INVARIANCE
static <T> void move(Stack<T>, from, Stack<T> to) {
    while(!from.isEmpty()) to.push(from.pop());
}
var rectS = new Stack<Rectangle>();
var graphS = new Stack<Graphic>();
graphS.push(new Rectangle()); // ok
move(rectS, graphS); // error
Stack<Object> objS = graphS; // error
/* anotherone */
String[] strA = new String[10];
Object[] objA = strA; // ok
objA[0] = Integer.valueOf(2); // runtime err
```

```
TODO:
// compiles?
List<Graphic> list = new
ArrayList<Rectangle>();
// compiles?
Rectangle[] rectangleStack = new Rectangle[10];
Graphic[] graphicStack = new Graphic[10];
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new
Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new
Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
for (Rectangle r : rectangleStack) {
    graphicStack.push(r);
}
```

```
COVARIANCE
public class Stack<E> {
    public void pushAll(Iterable<? extends E> src) { }
}
Stack<? extends Graphic> graphS;
graphS = new Stack<Rectangle>(); // ok
graphS = new Stack<Circle>(); // ok
graphS = new Stack<Object>(); // error (duh)
/* read only */
Stack<? extends Graphic> stack = new
Stack<Rectangle>();
stack.push(new Graphic()); // error
stack.push(new Rectangle()); // error
stack.push(new Triangle()); // error
```

```
CONTRAVARIANCE
static <T> void addToC(List<? super T> list, T e) {
    list.add(e);
}
```

```
addToC(new ArrayList<Number>(), 3); // ok
addToC(new ArrayList<Object>(), 3); // ok
/* shenanigans */
Stack<? super Graphic> s = new Stack<Object>();
s.add(new Graphic()); // ok
s.add(new Rectangle()); // ok
Graphic g = s.pop(); // error
Object g = s.pop(); // ok
```

```
BIVARIANCE
static void printList(List<?> list) {
    for (Object elem : list)
        System.out.print(elem + " ");
}
/* more shenanigans */
static void appendNewObject(List<?> list) {
    list.add(new Object()); // error
}
```

```
PRODUCER / CONSUMER
TODO:
```

```
MISC
<T extends Comparable & Collection> // multiple type
bounds
```

```
GENERICUS STREAM TOMFOOLERY
List<? extends Media> mediaList;
public <S extends T> List<S>
filterBySubtype(Class<S> subtype) {
    return mediaList.stream()
        .filter(subtype::isInstance)
        .map(subtype::cast)
        .collect(Collectors.toList());
}
```

```
ANNOTATIONS
- Metadata
- Not actually part of the code
```

```
TODO:
slides 10
```

```
- @Override
- @Test
- @Json...
```

```
DEFINING
@Target(ElementType.TYPE)
@Retention(RetentionPolicy.RUNTIME)
public @interface Author {
    String name();
    String date() default "N/A";
}
```

```
@Author(name = "UwU", date = "idon't validate input")
public class WeeOoo { }
```

```
REFLECTION
- Information of Class (private and public): Methods, Attrib-
utes, Parent class, Implemented interfaces
- Calling methods possible
- Usages: Debugger, Serialization, Frameworks, Remote
Procedure Call, ORM
```

```
// all of these `throws SecurityException`
public Field[] getDeclaredFields()
public Method[] getDeclaredMethods()
public Constructor<>[] getDeclaredConstructors()
```

```
System.out.println(dump(new Student("a", "b"), 0));
// {
//     firstName = a
//     lastName = b
// }
Class c = "s".getClass(); // java.lang.String
```

```
TODO:
Diagram (slides 26)
```

```
METHOD
// @Target(ElementType.METHOD)
public String getName()
public boolean isAnnotationPresent()
```

```
Class<? extends Annotation> annotationClass)
public Object invoke(Object obj, Object... args)
throws IllegalAccessException,
    IllegalArgumentException,
    InvocationTargetException
```

```
public class Profiler {
    public static void main(String[] args) {
        var testFunctions = new ProfileFunctions();
        int[] array = {5, 2, 8, 1, 93, 3, 33, 1, 333};
        var methods = ProfileFunctions.class
            .getDeclaredMethods();
        for (var m : methods) {
            if(m.isAnnotationPresent(Profile.class)) {
                m.setAccessible(true); // best practice or
                sth, idk, i write code in actually good languages
                Profiler.profileMethod(testFunctions, m,
                    new Object[] {array});
            }
        }
    }
}
```

```
CLASS
TODO:
```

```
// @Target(ElementType.TYPE)
getDeclaredConstructor().newInstance()
...
```

```
ATTRIBUTE
TODO:
```

```
// @Target(ElementType.FIELD)
...
```

```
VALIDATION
@Min(value = 18, message = "Age must be >= 18")
@Max(value = 99, message = "Age must be <= 99")
private int age;
@NotNull(message = "Name cannot be null")
private String name;
```

```
ALGORITHMS
- Brute-force
- Greedy
    - Take best option each step (optimize locally)
    - Fast
- Backtracking
    - Trial and error
    - Best overall option
    - Higher compute costs
- Divide and conquer
    - Binary search
- Dynamic programming
    - Recursion optimization?
    - Tabulation
```

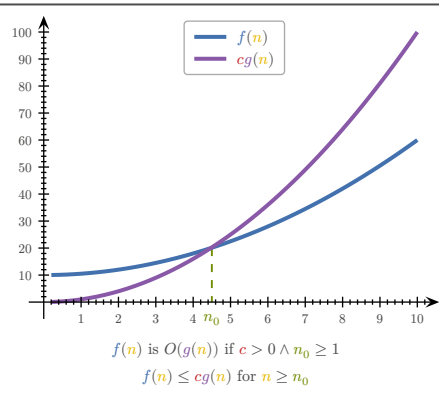
```
BACKTRACKING
TODO:
slides 87
```

```
BIG O NOTATION
- Worst case scenario
- Atomic operations = constant time
- Runtime measured as sum of primitive operations
```

Notation

$$\text{Algorithm } f \left(\underbrace{n}_{\text{Size of problem}} \right) = \text{Number of steps}$$

```
g = complexity class
TODO:
diagram  $f \leq cg$ 
```



f in order g:

Let $f, g: \mathbb{N} \rightarrow \mathbb{N}$
 $n_0, c \in \mathbb{N} \wedge \forall n \geq n_0: f(n) \leq c \cdot g(n)$
 $\Rightarrow f \in O(g) \Leftrightarrow f = O(g)$

Big O

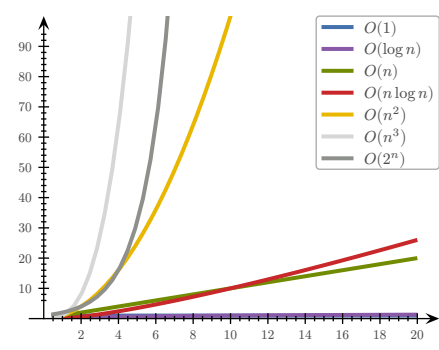
$$\widetilde{O} \left(\underbrace{n}_{\text{Size of problem}} \right) = \text{Complexity class}$$

RULES
If $f(n)$ is polynomial of degree d , then $f(n) \in O(n^d)$
- ignore lower powers
- ignore constants

Use most optimal function (lowest power)
- $2n$ is $O(n)$ better than $2n$ is $O(n^2)$

Simplify as far as possible
- $3n + 5$ is $O(n)$ instead of $3n + 5$ is $O(3n)$

COMPLEXITY CLASSES		
Name	Class	Example
Constant	1	Array read
Logarithmic	$\log n$	Binary search
Linear	n	Linear search
Log-linear	$n \log n$	Merge+Quick sort
Quadratic	n^2	Bubble+Insertion sort
Cubic	n^3	Solve equation
Polynomial	n^k	Various algorithms
Exponential	2^n	Calculate all subsets
Factorial	$n!$	Calculate all permutations



```
TODO:
Counting primitive operations (slides 55)
```

SORTING ALGORITHMS
SELECTION SORT
- Search for smallest elem in unsorted part and move in front

- Less mutations in array
- Same performance even if list almost sorted

```
public static void selectionSort(int[] a) {
    int n = a.length;
    for (int i = 0; i < n - 1; i++) {
        int minimum = i;
        for (int j = i + 1; j < n; j++)
            if (a[j] < a[minimum])
                minimum = j;
        swap(a, i, minimum);
    }
}
```

$$(n-1) + (n-2) + \dots + 1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2-n}{2} \sim O(n^2)$$

INSERTION SORT

- Take elem and put into correct place in sorted part
- Good for already partially sorted arrays

```
public static void insertionSort(int[] a) {
    int n = a.length;
    for (int i = 1; i < n; i++)
        for (int j = i; j > 0 && a[j] < a[j - 1]; j--)
            swap(a, j, j - 1);
}
```

$$1 + \dots + (n-1) + n = \sum_{i=1}^n i = \frac{n(n+1)}{2} = \frac{n^2+n}{2} \sim O(n^2)$$

BUBBLE SORT

- Compare neighboring elems and swap if needed

```
public static void bubbleSort(int[] a) {
    for (n = a.length; n > 1; n--)
        for (i = 0; i < n - 1; i++)
            if (a[i] > a[i + 1])
                swap(a, i, i + 1);
}
```

$$(n-1) + (n-2) + \dots + 1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2-n}{2} \sim O(n^2)$$

COUNTING SORT

```
public static void countingSort(int[] a) {
    int k = Arrays.stream(arr).max().getAsInt();
    int[] c = new int[k + 1];
    for (int i : a) c[i] += 1;
    int pos = 0;
    for (int i = 0; i < k; i++)
        for (int j = 0; j < c[i]; j++) {
            a[pos] = i;
            pos++;
        }
}
```

$$n + n + k + n = 3n + k \sim O(n + k)$$

SHELL SORT

- Sort pairs of far away elems
- Sort the partially sorted array through insertion sort

```
public static void shellSort(int[] a) {
    int n = a.length;
    int h = 1;
    while (h < n/3) h = 3 * h + 1;
    while (h >= 1) {
        for (int i = h; i < n; i++)
            for (int j = i; j >= h && a[j] < a[j - h]; j = j - h) swap(a, j, j - h);
        h /= 3;
    }
}
```

BINARY SEARCH

```
public static <T extends Comparable<T>> boolean bs(
    List<T> data, T target, int low, int high) {
    if (low > high) return false;
    int pivot = low + ((high - low) / 2);
    if (target.equals(data.get(pivot)))
        return true;
    else if (target.compareTo(data.get(pivot)) < 0)
        return searchBinary(data, target, low, pivot - 1);
    else
        return searchBinary(data, target, pivot + 1, high);
}
```

```
return searchBinary(data, target, pivot + 1, high);
}
```

$$O(\log n)$$

RECURSION

When a function calls itself

LINEAR RECURSION

Recursive call starts **at most one** further recursive call

TODO:
recursive → iterative (slides 61)

TAIL RECURSION

Linear recursion where the recursive call is **last**

TODO:
recursive → iterative (slides 69 – nice)

BINARY RECURSION

All non-terminal calls have **two** recursive calls

TODO:
example (slides 76,77,78)

TODO:
 $O(n)$ (slides 84)

DATA STRUCTURES

ADT: Abstract Data Type (e.g. Interface)

TODO:
diagram

ARRAY



LINKED LIST



DOUBLY LINKED LIST



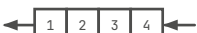
STACK

- LIFO



QUEUE

- FIFO



TREE

